

Symbolic and Evolutionary Artificial Intelligence

Student Group Project Guidelines

Marco Cococcioni

20 April 2026

A few words about the examination

Each examination session ("appello", in italian) is made of two parts:

- Practical examination ("prova pratica", in italian)
- Oral examination ("prova orale, in italian)

Each of the two parts (practical and oral) has its own date, typically scheduled 5 to 7 days apart. You must register **for each one** individually, as it is possible to take the oral examination in a later examination session, provided that you have passed the practical examination.

Practical examination ("presentation day")

Concerning the practical examination, students must form groups of **two or three**, and then *prepare a project* (more details below).

- During the practical examination each group will present his project (20 minutes). Then the teacher will ask clarification.
- If the project is approved, students may register individually for (and attend) the first available oral examination.
- If the project is **not approved**, revisions will be requested, and the group must resubmit the project again during a subsequent practical examination session (and give the presentation another time, of course).

Oral examination

Once a student has been admitted to the oral examination (having his project received the approval), he can register to the oral examination of the same session or any subsequent session.

Oral examinations will cover all the three main parts of the course (evolutionary multi-objective optimization, reinforcement learning, and numerical computation with infinitesimal and infinite numbers).

About the project

There are two types of projects:

- Standard RL Project (SRLP), on reinforcement learning
- Advanced Research Project (ARP), on an advanced topic agreed with the teacher.

Students must choose either one project type or the other; the two options are mutually exclusive.

Each project must be carried out by a group of **two or three** students.

Project types: Two Options

1. Standard RL Project (SRLP)

- Any project related to Reinforcement Learning (RL)
- Focus on benchmarking and implementation.
- Comparative analysis of known algorithms.
- Deliverable: 20-slide presentation + Code.

2. Advanced Research Project (ARP)

- Focus on original research or niche applications.
- Deep theoretical and experimental exploration.
- Deliverable: 20–30 page report (in A4 portrait format) + Code.

Comparison of Engagement Models

The primary difference between the two tracks lies in the autonomy required and the frequency of interaction with the teaching staff.

Feature	Standard RL Projects	Advanced Research Projects
Supervision	Low Interaction	Strong Interaction
Topic	Pre-approved / Common Gym	Custom / Teacher-agreed
Focus	Empirical Comparison	Scientific Contribution
Code	Extension/fork of existing repos	GitHub repo (+ Colab notebook if possible)
Main Goal	Practical proficiency	Research competence

- **Note on Standard RL Projects:** Students are expected to work with high autonomy. Guidance is provided during lab hours, but the design and execution are student-led.
- **Note on Advanced Research Projects:** Regular meetings will be scheduled to discuss progress, troubleshoot math/theory, and refine the methodology.

Standard RL Projects (SRLPs)

SRPs (1/4): Scope and Description

The project requires students (groups of 2–3) to design, implement, and evaluate an RL agent on a specific control or decision-making task.

- **Problem Selection:** Environments from OpenAI Gymnasium (e.g., *Cart-Pole*, *Lunar Lander*, *Frozen Lake*) or custom-built MDPs.
- **Core Task:** Implementation and rigorous comparison of at least **two distinct RL techniques** (e.g., DQN vs. PPO, or Q-Learning vs. SARSA).
- **Open Source Policy:** Forking existing GitHub repositories is allowed, provided students:
 1. Formally cite the original repository.
 2. **Extend** the codebase significantly (e.g., custom wrappers, different network architectures, or advanced optimizers).
 3. Prepare a detailed 20-slide presentation justifying the added value.

SRPs (2/4): Domain Analysis & MDP Modelling

Master's level rigor requires a "under the hood" understanding of the environment's dynamics.

Key Analytical Requirements

- **State Space:** Dimensionality analysis; Discrete vs. Continuous; Discussion on normalization or *frame stacking* requirements.
- **Action Space:** Impact of action types (Discrete, Continuous, Multi-discrete) on the choice of the algorithm.
- **Reward Engineering:** Analysis of the reward function structure (Sparse vs. Dense). Identification of potential *reward hacking* risks.
- **Stochasticity:** Evaluation of transition probabilities and environmental noise.

SRPs (3/4): Implementation & Technical Mastery

Students must demonstrate full control over the training pipeline and hyperparameter selection.

- **Algorithm Rationale:** Deep grasp of the mathematical foundations of the chosen methods (Value-based vs. Policy-gradient).
- **Training Dynamics:**
 - Justification for the number of episodes and convergence criteria.
 - Management of the **Exploration-Exploitation** trade-off.
 - Usage of **Replay Buffers**: Sizing, sampling strategies, and impact on sample efficiency.
- **Architecture:** Motivations for the Neural Network topology, activation functions, and the use of *Target Networks* for stability.

SRPs (4/4): Evaluation, Performance & Engineering Metrics

As AI & Data Engineers, students must go beyond qualitative observation.

- **Statistical Significance:** Results must be averaged over multiple seeds. Use of confidence intervals and statistical tests (e.g., t-test) to validate the comparison.
- **Efficiency Analysis:**
 - **Training Time:** Analysis of sample complexity.
 - **Inference Time:** Latency evaluation for real-time deployment.
- **Generalisation:** Evaluation of the agent on variations of the initial environment.
- **Failure Analysis:** Critical discussion on where the agent fails and the limitations of the chosen algorithms (e.g., *brittleness* or *catastrophic forgetting*).

Advanced Research Projects (ARPs)

ARPs (1/4): Scope and Description

The Advanced Research Project is designed for students seeking to explore cutting-edge research topics (**a list is provided below**).

- **Custom Research Topic:** The specific problem or research question must be discussed and formally agreed upon with the teacher.
- **Teacher Interaction:** High frequency of interaction is expected. Students act as junior researchers, receiving guidance on methodology, literature review, and experimental design.
- **Innovative Nature:** Unlike standard reinforcement learning projects, this track rewards original contributions on difficult, open problems.

ARPs (2/4): Technical Deliverables & Workflow

Rigour in version control and reproducibility is a core requirement for this type of project.

- **Codebase:** The preferred format is a **Python Jupyter Notebook**.
- **Accessibility:**
 - Code must be hosted in a private **GitHub** repository.
 - Access must be shared with the teacher.
 - Integration with **Google Colab** is highly encouraged to facilitate interactive debugging and review.
- **Reproducibility:** The notebook must be self-contained, including requirements, environment setup, and saved model weights for inference testing.

ARPs (3/4): The Research Report

A formal scientific report is the primary deliverable, demonstrating the student's ability to communicate complex technical findings.

- **Format:** A4 document, typically ranging from **20 to 30 pages**.
- **Structure:**
 1. **Abstract & Introduction:** Motivation and problem statement.
 2. **Literature Review:** Positioning the work within the current state-of-the-art RL research.
 3. **Methodology:** Detailed description of algorithms, network architectures, and MDP modelling.
 4. **Experimental Setup:** Hardware, software, and hyperparameter tuning.
 5. **Discussion:** Critical analysis of results, limitations, and future work.

ARPs (4/4): Assessment Criteria

The evaluation focuses on research autonomy and technical depth.

The teacher looks for:

- **Analytical Depth:** Beyond "running code"—why did you make this choice here? What is the role of this portion of the code? Why did you use this loss function and not this other one? Why didn't you try to regularize the neural network?
- **Literature Mastery:** Ability to cite and compare results with existing academic papers.
- **Scientific Communication:** Clarity of the report, quality of visualisations (plots, heatmaps, saliency maps), and precision of the technical terms used.

ARPs: Possible Research Topics

ARPs: Possible Research Topics

- Photonic Neural Networks
- Formal Verification of properties of Neural Networks
- Privacy Preserving Neural Networks (deals with the use of encrypted weights and cryptography)
- Quantum Optimization (single and multi-objective) and Quantum Neural Networks
- Prompt-based Image Generation using Diffusion Models for Data Augmentation
- Experiments with Graph Neural Networks in the Costruction Industry (Building Information Modelling)
- Implementing DNN kernel acceleration on GPUs (using CUDA C)
- Implementing the inference of a DNN on a Field Programmable Gate Array (FPGA)
- Implementing and testing Agentic AI (LangChain and LangGraph) (**only for those of you that want to do their Master Thesis on this topic**)

How to Prepare the Documentation

Each group must prepare the documentation described above using latex:

- 20-slide **presentation** (for SRLPs)
- 20-30 page **report** (for ARPs)

During the Practical examination, they will present the associated PDF file.

How to Prepare the Documentation (2/2)

- The presentation (or the report) must be generated using **overleaf**.
- Use the University of Pisa overleaf subscription (<https://tex.dm.unipi.it/>). Log in using your ...@studenti.unipi.it account
- Give the following name to the project:
2026_SEAI_project_Lastname1_Lastname2(_Lastname3)
- Share the project with the teacher (use this overleaf email: marco.cococcioni@unipi.it) **at least 3 days** before the practical examination date (the "*presentation day*").
- You can share the project even earlier (ideally, right after creating it). This allows the teacher to review your work while it is in progress and, whenever possible, enables you to get feedback from the teacher even before the *presentation day*.